

CALCULUS OF FINANCIAL PATHS

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► **Example:** for any ϕ -divergence with twice-differentiable convex ϕ with $\phi(1) = \phi'(1) = 0$

$$D_f(\pi_{\beta'}||\pi_{\beta})=\pi_{\beta}\left[\phi(1+\Delta\pi_{\beta,\beta'})\right]$$

The product rule is very powerful and allows to deal with paths

$$\frac{\mathrm{d}^2}{\mathrm{d}\beta^2}\pi_\beta[f]=\pi_\beta[f(\dot{\pi}_\beta^2+\ddot{\pi}_\beta)]$$

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\pi_{\beta}[f_{\beta}]=\dot{\pi}_{\beta}[f_{\beta}]+\pi_{\beta}[\dot{f}_{\beta}]$$

► **Product rule:** If $f_{\beta}(x)$ is differentiable in β for all x with derivative $\dot{f}_{\beta}(x)$ with respect to β :

► **Example:** take higher order derivatives in terms of the derivatives of f_β and $\dot{\pi}_\beta$. For example

$$= \frac{1}{2} \phi''(1) \Delta \beta^2 I(\beta) + O(\Delta \beta^3)$$

► For example if f is constant in β ,

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$$\frac{d}{d\beta} \pi_\beta[f_\beta] = \dot{\pi}_\beta[f_\beta] + \pi_\beta[\dot{f}_\beta]$$

- ▶ The product rule is very powerful and allows us to do calculus on paths
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$$\frac{d^2}{d\beta^2} \pi_\beta[f] = \pi_\beta[f(\dot{\pi}_\beta^2 + \ddot{\pi}_\beta)]$$

- ▶ **Example:** for any ϕ -divergence with twice-differentiable convex ϕ with $\phi(1) = \phi'(1) = 0$

$$\begin{aligned} D_f(\pi_{\beta'} \| \pi_\beta) &= \pi_\beta \left[\phi(1 + \Delta\pi_{\beta,\beta'}) \right] \\ &= \frac{1}{2} \phi''(1) \Delta\beta^2 I(\beta) + O(\Delta\beta^3) \end{aligned}$$

STEIN VS FISHER SCORE